

Conservation of energy

The first law of thermodynamics (the study of how heat behaves) states that energy cannot be created or destroyed, but it can be transferred from one object to another and converted into different forms.

some of the rider's energy is converted to thermal energy, making him hot

climbing the hill, the rider's kinetic energy is converted into potential energy, which will be released (converted back into kinetic energy) when the rider freewheels down the other side

chemical energy in the rider's muscles makes the legs move

kinetic energy is transferred via the pedals and the chain to the back wheel

some of the wheels' kinetic energy becomes thermal energy, heating the bicycle's tires as they rub against the ground

△ Energetic bicycling

The person is making the bicycle move by pushing on the pedals. As the bicycle goes faster, it gains kinetic energy. This is possible due to the chemical energy released in the rider's muscles that makes the legs move. At some point, the rider can use this same chemical energy in the muscles to stop the bike.

All machines will gradually lose energy, which, unfortunately, makes **perpetual motion impossible**.

REAL WORLD

Perpetual motion

For many years inventors have tried to develop a machine that could run forever. This machine shown right, designed by the German Ulrich von Cranach in 1664, was driven by cannonballs falling into the large wheel at the right. These would drop onto a curved track that fed them into an Archimedes screw. Powered by the wheel, the screw lifted the balls to the starting position. However, like all perpetual motion machines before and since, this clever design could not overcome the slowing effect of friction (see page 173).

